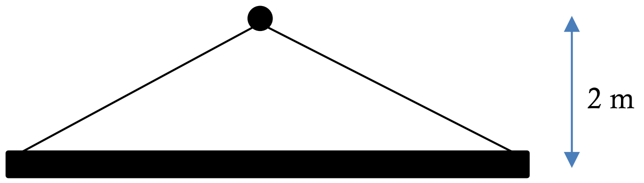
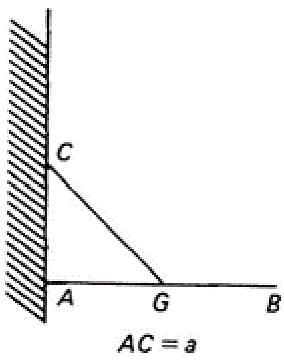
THREE FORCE PROBLEMS **22/04/26**

1. A uniform beam of mass 8 kg and length 6 m is suspended horizontally by two *separate* strings hooked to a fixed hook, which is 2 m vertically above the line of the beam.
    a. Use the three force theorem to explain why the hook must be vertically above the centre of the beam.
    b. Use a triangle of forces to find the tensions in the two strings.



2. A uniform rod AB of length $2a$ and weight W, smoothly pivoted at A, is held in equilibrium by a string. G is the midpoint of AB. Use force triangles and the Three-Force theorem to help find
    a. the tension in the string
    b. the magnitude and direction of the reaction at A

i.



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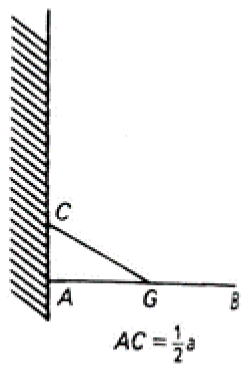

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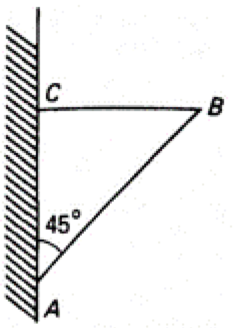

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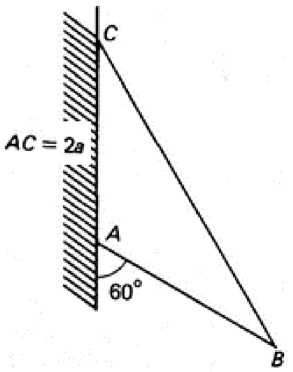

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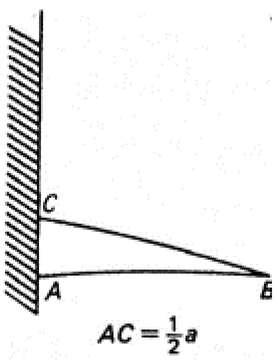

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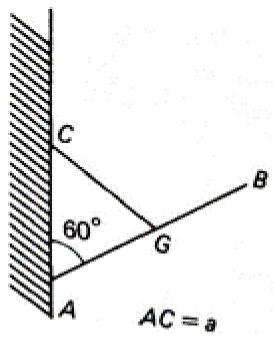
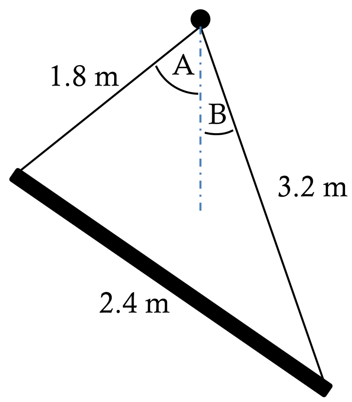


3. A uniform beam of length 2.4 m and mass 4 kg is suspended from its ends by separate strings of lengths 1.8 m and 3.2 m respectively. The other ends of the strings are attached to a fixed point, and the beam hangs in equilibrium.
   a. Find the angles marked A and B.
   b. Find the angle the beam makes with the horizontal.
   c. Find the tensions in the two strings.



4. A uniform rectangular lamina ABCD of weight W, with AB = $8a$ and AD = $6a$, is smoothly pivoted to a fixed point at A. It is kept in equilibrium by in a vertical plane with AC horizontal and B as the lowest point by a force P. Find the magnitude of P if:
   a. P acts along EF, where E and F are the midpoints of AB and CD
   b. P acts along BC
   c. P acts horizontally, and the force exerted by the pivot acts in the direction BA.